

Name: _____

Date: _____

Year 11 Advanced Mathematics — Calculus Review

Limits, Differentiation and Integration

Learning Intentions

By the end of this worksheet you will be able to:

- Evaluate limits using algebraic techniques
- Differentiate polynomial, power, and composite functions
- Apply integration as the reverse of differentiation
- Identify and correct common calculus errors

Section A: Fluency

1. Evaluate each limit. Show any algebraic manipulation required.

- (a) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$
- (b) $\lim_{x \rightarrow 0} \frac{5x^3 - 2x}{x}$
- (c) $\lim_{x \rightarrow \infty} \frac{4x^2 + 1}{2x^2 - 3}$

2. Differentiate each function with respect to x . Simplify your answer fully.

- (a) $f(x) = 3x^4 - 7x^2 + 2x - 5$
- (b) $g(x) = \frac{4}{x^3} + \sqrt{x}$
- (c) $h(x) = (2x - 1)^5$

Show working clearly

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5. A student evaluates $\int_1^3 (2x + 1) \, dx$ as follows:

Student's Working:

$$\begin{aligned} \int_1^3 (2x + 1) \, dx &= [x^2 + x + C]_1^3 \\ &= (9 + 3 + C) - (1 + 1 + C) \\ &= 12 + C - 2 - C \\ &= 10 \end{aligned}$$

The answer is $10 + C$.

- (a) The student's final numerical value of 10 is correct, but they have made a conceptual error in their final statement. Identify and explain it.
- (b) State the correct final answer and explain why C does not appear in a definite integral result.

Section C: Investigation Skills

Section C Instructions

These questions ask you to explore relationships, identify variables, and draw evidence-based conclusions. Show all reasoning clearly.

6. Investigation: Gradient of a Curve

Consider the function $f(x) = x^3 - 3x$.

- (a) Find $f'(x)$.
- (b) Complete the table below by evaluating $f'(x)$ at each value of x .

x	-2	-1	0	1	2
$f'(x)$					

Show working clearly

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3. Find each indefinite integral. Include the constant of integration.

- (a) $\int (6x^2 - 4x + 3) \, dx$
- (b) $\int x^{-4} \, dx$
- (c) $\int (3x + 1)^2 \, dx$

Section B: Spot the Error

Section B Instructions

Each question below contains exactly **one error** in the student's working. For each question:

- Identify the **line** where the error occurs.
- State **what the error is** and why it is incorrect.
- Provide the **correct solution**.

4. A student differentiates $f(x) = (x^2 + 3)^4$ as follows:

Student's Working:

$$\begin{aligned} f(x) &= (x^2 + 3)^4 \\ f'(x) &= 4(x^2 + 3)^3 \end{aligned}$$

- (a) On which line does the error occur? What rule has the student forgotten to apply?
- (b) Write the correct derivative, showing full working.

Show working clearly

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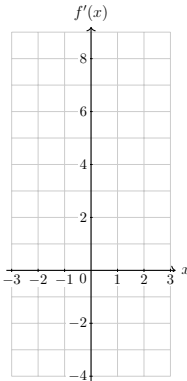
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(c) Using your table, identify the x -values where the gradient is zero. What do these points represent on the curve?

(d) **Independent variable (IV):** What are you changing in this investigation?

Dependent variable (DV): What are you measuring?

(e) Sketch $f'(x)$ on the axes below. Label the x -intercepts.



(f) Write one evidence-based conclusion linking the sign of $f'(x)$ to the behaviour (increasing or decreasing) of $f(x)$.

Show working clearly

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Year 11 Mathematics Advanced – Calculus Review

Teacher-review draft aligned to Stage 6 introductory differential calculus.

Learning Intentions

- Differentiate polynomial and simple power functions.
- Interpret gradients and rates of change.
- Identify and correct common differentiation errors.
- Investigate how changing parameters affects gradients.

1. Differentiate the following functions.

- (a) $f(x) = 3x^4 - 2x^2 + 7$
(b) $g(x) = 5x^3 + 4x - 9$
(c) $h(x) = \frac{1}{2}x^5 - 3x^2$

2. For $y = x^3 - 6x$:

- (a) Find $\frac{dy}{dx}$.
(b) Determine the gradient when $x = 2$.
(c) State whether the function is increasing or decreasing at $x = 2$.

Show working clearly

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3. Spot the Error

A student differentiates $y = 4x^3 - 7x^2 + 2$ as:

$$\frac{dy}{dx} = 12x^3 - 14x.$$

- (a) Identify the exact error.
(b) Write the correct derivative.
(c) Explain why the student's method is incorrect.

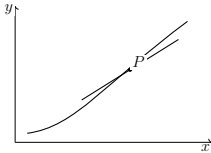
4. Spot the Error

A student claims:

$$\frac{d}{dx}(8) = 8.$$

- (a) Explain the misconception.
(b) State the correct derivative.
(c) Give another example involving a constant.

5. The graph below represents a function at a point P .



Show working clearly

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Describe what the gradient of the tangent at P tells us about the behaviour of the function at that point.

6. Investigation Skills

Consider the family of functions:

$$y = ax^2.$$

Complete the table.

a	Derivative	Gradient at $x = 2$
1		
2		
5		

What pattern do you notice as a increases?

7. Investigation Skills

A student proposes the hypothesis:

“Doubling every coefficient in a polynomial doubles the gradient at every value of x . ”

Test the hypothesis using the function

$$f(x) = x^3 - 2x + 1.$$

Show evidence and state whether the hypothesis is supported.

Show working clearly

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8. A particle moves so that its displacement is modelled by

$$s = t^3 - 3t^2 + 2t,$$

where s is measured in m and t in s.

- (a) Determine the velocity function.
(b) Find the velocity when $t = 1$.
(c) Explain the meaning of your answer in context.

Show working clearly

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Year 11 Advanced Mathematics — Calculus Review

Learning intentions (teacher-review draft)

- Review differentiation of polynomials, simple products, and quotients.
- Interpret derivatives graphically and in context.
- Identify and correct common calculus misconceptions.
- Apply investigation skills: variables, data, reasoning, and conclusions.

1. Differentiate each function. Show all working.

(a) $f(x) = 3x^4 - 5x^2 + 7$

(b) $g(x) = \frac{2}{x^3}$

(c) $h(x) = (x^2 + 1)(3x - 4)$

Show working clearly

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4. Investigation: Average vs Instantaneous Rate of Change.

A particle moves along a line with position function $s(t) = t^3 - 6t^2 + 9t$ (where s is in m and t in s).

(a) Identify the independent and dependent variables.

(b) Complete the data table for average rate of change over the interval $[1, 1 + h]$.

h (s)	Average rate of change (m/s)
0.5	
0.1	
0.01	

(c) Use your table to estimate $s'(1)$.

(d) Explain how the data supports the concept of instantaneous rate of change.

5. A function $p(x)$ has derivative $p'(x) = 6x - 4$.

(a) Find the x -coordinate of any stationary point(s).

(b) Determine whether the stationary point is a maximum, minimum, or neither. Justify your answer.

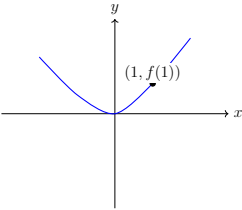
(c) Sketch a possible graph of $p'(x)$ and annotate key features.

Show working clearly

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2. The graph below shows a sketch of $y = f(x)$. Estimate $f'(x)$ at $x = 1$ and explain your reasoning.



3. Spot the Error. A student differentiates $k(x) = \frac{x^2 - 3}{x}$ as follows:

$$k'(x) = \frac{2x}{x} - \frac{x^2 - 3}{x^2}.$$

(a) Identify the exact step where the error occurs.

(b) Correct the working.

(c) Explain why the student's method fails in general.

Show working clearly

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Teacher Answer Key

6. (a) $f'(x) = 12x^3 - 10x$

(b) $g(x) = 2x^{-3} \Rightarrow g'(x) = -6x^{-4}$

(c) Product rule:

$$h'(x) = 2x(3x - 4) + (x^2 + 1)(3) = 6x^2 - 8x + 3x^2 + 3 = 9x^2 - 8x + 3.$$

7. Slope of tangent at $x = 1$ is approximated using a secant. From the sketch,

$$\frac{f(1.2) - f(1)}{1.2 - 1} \approx \frac{1.2 - 0.8}{0.2} = 2.$$

Any estimate between 1.5 and 2.5 with justification is acceptable.

8. (a) Error: treating $\frac{x^2 - 3}{x}$ as a quotient requiring the quotient rule but incorrectly applying it.

(b) Correct method: simplify first.

$$k(x) = x - \frac{3}{x} \Rightarrow k'(x) = 1 + 3x^{-2}.$$

(c) Explanation: The student attempted a quotient rule but omitted the derivative of the denominator and mis-structured the rule.

9. (a) IV: t ; DV: s .

(b) Average rates:

$$\frac{s(1 + h) - s(1)}{h}.$$

$$h = 0.5: 1.75 \text{ m/s}; h = 0.1: 1.09 \text{ m/s}; h = 0.01: 1.009 \text{ m/s}.$$

(c) Estimate: $s'(1) = 1$.

(d) As h decreases, the average rate approaches a limiting value, illustrating instantaneous rate.

10. (a) Solve $6x - 4 = 0 \Rightarrow x = \frac{2}{3}$.

(b) $p''(x) = 6 > 0$ so the stationary point is a minimum.

(c) Graph: straight line with slope 6, crossing the x -axis at $x = \frac{2}{3}$.

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Year 11 Mathematics Advanced: Calculus Review
Spot the Error + Investigation Skills

Teacher-review draft

This worksheet reviews introductory calculus ideas for Stage 6 Mathematics Advanced. It focuses on gradient, average rate of change, instantaneous rate of change, differentiation, tangent lines, and interpretation from data.

Learning intentions

By the end of this worksheet, you should be able to:

- connect gradient, rate of change, and the derivative;
- differentiate simple polynomial functions;
- find tangent and normal equations;
- identify and correct common calculus errors;
- use data to estimate and interpret rates of change.

Key idea

Average rate of change over $[a, b]$ is

$$\frac{f(b) - f(a)}{b - a}.$$

Instantaneous rate of change at $x = a$ is represented by $f'(a)$.

1. Access starter: average rate of change

A function is given by

$$f(x) = x^2 + 2x.$$

- (a) Find $f(1)$ and $f(4)$.
- (b) Find the average rate of change of $f(x)$ from $x = 1$ to $x = 4$.
- (c) Explain what the average rate of change represents on a graph.

Show working clearly

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4. Fluency: differentiate

Differentiate each function.

- (a) $y = 5x^4 - 3x^2 + 7$
- (b) $y = \frac{6}{x^2}$
- (c) $y = (2x - 1)(x + 4)$
- (d) $y = 3\sqrt{x}$

5. Spot the Error: power rule

A student differentiates

$$y = \frac{8}{x^3}$$

as follows:

$$y = 8x^3$$
$$\frac{dy}{dx} = 24x^2.$$

- (a) Identify the exact error.
- (b) Correct the working.
- (c) Explain the key rule the student forgot.

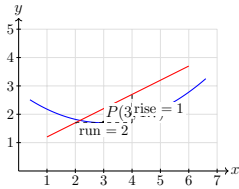
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2. Gradient from a graph

The diagram below shows a curve and a tangent line at point P .



- (a) Use the tangent triangle to estimate the gradient at P .
- (b) State the approximate value of $f'(3)$.
- (c) Explain why the tangent gradient is more useful than a secant gradient when describing instantaneous rate of change.

3. First principles

Use first principles to find the derivative of

$$f(x) = x^2 + 3x.$$

Start from

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Show working clearly

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6. Tangent and normal

The curve is

$$y = x^2 - 4x + 1.$$

- (a) Find $\frac{dy}{dx}$.
- (b) Find the gradient of the tangent at $x = 3$.
- (c) Find the coordinates of the point on the curve when $x = 3$.
- (d) Find the equation of the tangent at $x = 3$.
- (e) Find the equation of the normal at $x = 3$.

7. Investigation skills: motion from data

A ball rolls along a straight track. Its distance from the starting point is recorded below.

Time, t (s)	Distance, s (m)
0	0.0
1	0.7
2	2.8
3	6.3
4	11.2
5	17.5

- (a) Calculate the average velocity from $t = 1$ to $t = 4$.
- (b) Estimate the instantaneous velocity at $t = 3$ using the symmetric interval from $t = 2$ to $t = 4$.
- (c) State one controlled condition that should be kept constant during the experiment.
- (d) State one possible source of measurement error.
- (e) The data appears to follow $s = 0.7t^2$. Use calculus to find the velocity function.

Show working clearly

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Calculus: Spot the Error & Kinematic Investigation

Learning Intentions:

- Identify and correct algebraic errors in First Principles differentiation.
- Apply the power rule fluently to terms with negative and fractional indices.
- Investigate instantaneous rates of change in a physical context (kinematics).

1. Fluency: The Power Rule

Differentiate each of the following functions with respect to x . Express your final answers with positive indices where applicable.

(a) $y = 3x^4 - 5x^2 + 7x - 2$

(b) $f(x) = \frac{4}{x^3} - \sqrt{x}$

2. Diagnostic: Spot the Error

A student was asked to find the derivative of $f(x) = x^2 - 4x$ using differentiation from first principles. Their working is shown below:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 4x - (x^2 - 4x)}{h} \quad \dots\dots \text{(Line 1)} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + h^2 - 4x - x^2 + 4x}{h} \quad \dots\dots \text{(Line 2)} \\ &= \lim_{h \rightarrow 0} \frac{h^2}{h} \\ &= \lim_{h \rightarrow 0} h = 0 \end{aligned}$$

Show working clearly

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- (a) Identify and explain the **two** distinct mathematical errors the student made in Lines 1 and 2.

- (b) Provide the correct full working to find $f'(x)$ from first principles.

3. Investigation Skills: Motorcycle Kinematics

A motorcycle accelerates away from a set of traffic lights along a straight road. Its displacement, s metres, from the traffic lights after t seconds is modelled by the function:

$s(t) = 2t^3 - 15t^2 + 36t \quad \text{for } t \geq 0$

- (a) Determine the velocity function, $v(t)$, of the motorcycle.

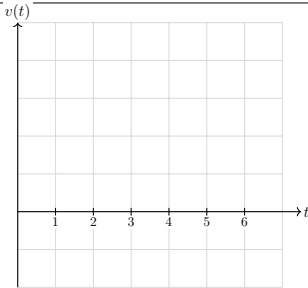
- (b) Investigate the times at which the motorcycle is instantaneously at rest.

- (c) Construct a sign table to investigate the intervals where the velocity is positive (moving forwards) and negative (moving backwards).

- (d) On the grid below, sketch the velocity-time graph to support your investigation findings from parts (b) and (c).

Show working clearly

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4. Synthesis: Parameter Shift

The curve $y = ax^2 + bx$ has a tangent with a gradient of 4 at the point (1,3). By investigating the properties of the gradient function and the given coordinates, set up simultaneous equations to find the exact values of the constants a and b .

Show working clearly

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Teacher Answer Key & Marking Guide

1. Fluency: The Power Rule

(a) $y' = 12x^3 - 10x + 7$ [1 mark for each correct term]

(b) $f(x) = 4x^{-3} - x^{1/2}$ (Index form setup)
 $f'(x) = -12x^{-4} - \frac{1}{2}x^{-1/2} = -\frac{12}{x^4} - \frac{1}{2\sqrt{x}}$ [1 mark for negative index rule, 1 mark for fractional index rule]

2. Diagnostic: Spot the Error

- (a) **Error 1 (Line 1):** The student failed to substitute $(x + h)$ into the $-4x$ term. It should read $-4(x + h)$, not just $-4x$.
Error 2 (Line 2): The student incorrectly expanded the perfect square $(x + h)^2$ as $x^2 + h^2$. They missed the middle term $2xh$. [1 mark for each correctly identified error]

(b) Correct working:
$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 4(x+h) - (x^2 - 4x)}{h} \\ f'(x) &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 4x - 4h - x^2 + 4x}{h} \\ f'(x) &= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 4h}{h} = \lim_{h \rightarrow 0} (2x + h - 4) = 2x - 4 \end{aligned}$$
 [2 marks for correct expansion and limit evaluation]

3. Investigation Skills: Motorcycle Kinematics

(a) $v(t) = s'(t) = 6t^2 - 30t + 36$ [1 mark]

- (b) Set $v(t) = 0$:
 $6(t^2 - 5t + 6) = 0 \implies 6(t - 2)(t - 3) = 0$
The motorcycle is at rest at $t = 2$ and $t = 3$ seconds. [1 mark factorisation, 1 mark solutions]

- (c) Sign table:
 $t = 1 \implies v(1) = 6(1)^2 - 30(1) + 36 = 12$ (+)
 $t = 2.5 \implies v(2.5) = 6(-0.5)(0.5) = -1.5$ (-)
 $t = 4 \implies v(4) = 6(2)(1) = 12$ (+)
Moves forwards for $0 \leq t < 2$ and $t > 3$. Moves backwards for $2 < t < 3$. [2 marks for valid table/test points]

- (d) Graph should be a concave-up parabola. t -intercepts at $t = 2$ and $t = 3$. v -intercept at $v = 36$ (will be off the provided standard scale, look for correct shape and roots). [2 marks for roots and correct concavity]

4. Synthesis: Parameter Shift

Show working clearly

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Year 11 Calculus Review: Derivatives

Key Idea: The derivative measures the rate of change of a function. Use rules such as power, product, and chain rules.

1. Find the derivative of the following functions:

- (a) $f(x) = 3x^4 - 5x^2 + 7$
- (b) $g(x) = \frac{1}{x^2}$
- (c) $h(x) = (2x + 3)^5$

2. Find the gradient of the tangent to $y = x^3 - 4x$ at $x = 2$.

3. A particle moves along a line with position given by $s(t) = 2t^3 - 3t^2 + t$.

- (a) Find the velocity function.
- (b) Find the acceleration function.
- (c) Determine when the particle is at rest.

- (d) Verify your hypothesis using differentiation.

7. Application

The height of a ball is given by $h(t) = -5t^2 + 20t + 1$.

- (a) Find the velocity function.
- (b) Determine when the ball reaches maximum height.
- (c) Find the maximum height.

4. Spot the Error

A student differentiates $y = (x^2 + 1)^3$ as follows:

$\frac{dy}{dx} = 3(x^2 + 1)^2$

- (a) Identify the mistake.
- (b) Correct the derivative.
- (c) Explain why the chain rule is required.

5. Spot the Error

A student claims:

$\frac{d}{dx}(x \sin x) = \sin x$

- (a) Identify the mistake.
- (b) State the correct derivative.
- (c) Name the rule used.

6. Investigation Skills

Consider the function $f(x) = x^2$.

- (a) Complete the table of average rates of change between $x = 1$ and nearby points:

x	Average rate of change
1.5	
1.1	
1.01	

- (b) What value do these rates approach?
- (c) Form a hypothesis for the gradient at $x = 1$.

Teacher Answer Key

- 1. (a) $f'(x) = 12x^3 - 10x$
(b) $g'(x) = -2x^{-3} = -\frac{2}{x^3}$
(c) $h'(x) = 5(2x + 3)^4 \cdot 2 = 10(2x + 3)^4$

2. $\frac{dy}{dx} = 3x^2 - 4$
At $x = 2$: $3(4) - 4 = 8$

- 3. (a) $v(t) = 6t^2 - 6t + 1$
(b) $a(t) = 12t - 6$
(c) Solve $6t^2 - 6t + 1 = 0$

4. Error: chain rule missing inner derivative.
Correct: $6x(x^2 + 1)^2$

5. Error: product rule not used.
Correct: $\sin x + x \cos x$

6. Rates approach 2.
Derivative: $f'(x) = 2x$, so at $x = 1$, gradient = 2.

- 7. (a) $v(t) = -10t + 20$
(b) $-10t + 20 = 0 \Rightarrow t = 2$
(c) $h(2) = 21$

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Year 11 Mathematics Advanced
Introduction to Calculus: Stationary Points Investigation

Syllabus Focus: MA-C1 Introduction to Calculus

This worksheet reviews foundational differential calculus concepts, focusing on the formal definition of the derivative (first principles), the geometric interpretation of gradients, and algebraic differentiation rules.

1. Spot the Error (First Principles)

A student is asked to differentiate $f(x) = 3x^2 - 5x$ from first principles. Their working is shown below:

$$\begin{aligned} f'(x) &= \frac{f(x+h) - f(x)}{h} \\ &= \frac{3(x+h)^2 - 5x + h - (3x^2 - 5x)}{h} \quad \text{--- Step 1} \\ &= \frac{3(x^2 + 2xh + h^2) - 5x + h - 3x^2 + 5x}{h} \quad \text{--- Step 2} \\ &= \frac{3x^2 + 6xh + 3h^2 + h - 3x^2}{h} \quad \text{--- Step 3} \\ &= \frac{6xh + 3h^2 + h}{h} \quad \text{--- Step 4} \\ &= 6x + 3h + 1 \quad \text{--- Step 5} \\ &= 6x + 1 \quad \text{--- Step 6} \end{aligned}$$

Identify the **two** distinct algebraic or conceptual errors made by the student. Explain why each is incorrect and provide the completely corrected full working.

Show working clearly

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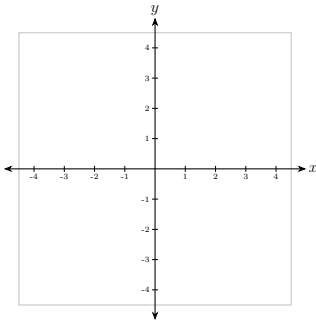
2. Geometric Investigation Skills: Curve Analysis & Stationary Points

Consider the curve defined by the function $f(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$.

- a. Determine the algebraic expression for the gradient function, $f'(x)$.
- b. Find the coordinates of the points on the curve where the tangent is completely horizontal (parallel to the x -axis).
- c. Construct a first-derivative sign table for the critical values found in part (b) to classify the nature of these stationary points.
- d. State the intervals of x for which the curve $y = f(x)$ is strictly increasing.
- e. Investigate the behavior of the gradient across the domain by sketching the gradient function $y = f'(x)$ on the grid below. Clearly label all intercepts.

Show working clearly

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Show working clearly

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3. Algebraic Mastery & Parameter Shift

Differentiate the following functions with respect to x . Express your answers with positive indices and in radical form where applicable.

- a. $f(x) = \frac{4}{x^3} - \sqrt{x}$
- b. Given that $g(x) = ax^n + bx^2$, where a and b are constants, it is known that $g'(1) = 10$ and $g'(2) = 32$. Determine the exact values of parameters a and n if n is a positive integer greater than 2.

Show working clearly

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